

Low BOM cost EDP to LVDS conversion solution CS5211

At present, the EDP to [LVDS](#) conversion solutions on the market include: Longxun LT8911 LT7211, Bridgelux PS8625, Quintiles CH7511 and other solutions. DP to LVDS conversion is mainly used in some [LCDs](#) with LVDS interfaces to transfer EDP signals to LVDS screen output. At this time, an adapter board such as EDP TO LVDS is needed to achieve screen display effects. Below we will focus on the characteristics of the above solutions, and also recommend a more cost-effective and low-cost EDP to LVDS conversion solution CS5211.

The Longxun LT8911 is an LVDS to eDP converter with a single-port or dual-port configurable LVDS receiver, with 1 clock channel and up to 10 data channels, each data channel has a maximum operating rate of 1.5Gbps; the maximum input bandwidth is 12Gbps. The converter deserializes the input LVDS data, decodes the data packet, and converts the formatted video data stream into a single-link output compliant with VESA eDP1.4, with 1/2/4 configurable data channels, supporting RBR (1.62Gbps) and HBR (2.7Gbps) link speeds. Built-in optional subsystem controller functions [can](#) reduce the impact of [EMI](#) on EMI-related system applications.

The Longxun LT7211 is a high performance C/DP1.2 to LVDS chip for VR/display applications. For DP1.2 input, the LT7211 can be configured as 1, 2, 4 lanes, and also supports lane [swapping](#) function. Adaptive equalization makes it suitable for long cable applications, and the maximum bandwidth can reach 21.6Gbps. For LVDS output, the LT7211 can be configured as single port, dual port, or quad port. For 2D video streams, the same video stream can be mapped to two separated panels, and for 3D video formats, the left data can be sent to one panel and the right data can be sent to another panel.

The Bridgelux PS8625 is a DisplayPort™ to LVDS converter designed for PCs utilizing a GPU and a DisplayPort (DP™) or Embedded DisplayPort (eDP™) output and a display panel that accepts LVDS input. The PS8625 will appear as a DP or eDP sink device to the video source and will act as an LVDS source device to an LVDS display panel. The device is a fully integrated solution requiring no external CPU, memory, clock reference or voltage regulators. The PS8625 can be configured to read EDID from the monitor DDC channel or from an optional external ROM connected to the PS8625 (provided an I2C host).

Quintiles CH7511B is an EDP to LVDS semiconductor device that converts embedded DisplayPort signals to LVDS (Low Voltage Differential Signaling). This innovative DisplayPort receiver with integrated LVDS transmitter is designed for the All-in-One PC and Notebook market. The CH7511B is designed to comply with the Embedded DisplayPort Specification Version 1.2. In the receiver module of the device, it supports two eDP Main Link inputs with data rates of 1.62Gb/s or 2.7Gb/s and can accept 18-bit 6:6:6 or 24-bit 8:8:8 RGB digital formats for LVDS outputs up to 1920x1200.

In summary: Longxun LT8911 LT7211, Bridgelux PS8625, Quintiles CH7511 and other solutions can all meet the requirements of EDP to LVDS conversion during the design process. The difference lies in the design of the peripheral circuit and the pinout of the chip. The advantage of Richida CS5211 EDP to LVDS Conversion Scheme is that it has fewer peripheral circuit devices and the chip cost is lower than that of Longxun LT8911 LT7211, Bridgelux PS8625, Quintiles CH7511. The compatibility of CS5211 is not worse than that of Longxun LT8911 LT7211, Bridgelux PS8625, Quintiles CH7511. The supply cycle of CS5211 is shorter, and the feasibility and stability of the solution are better than those of Longxun LT8911 LT7211, Bridgelux PS8625, Quintiles CH7511.

The CS5211 is an EDP to LVDS converter chip solution that converts eDP input to LVDS signal. The advanced receiver of the CS5211 supports the embedded display port (eDP) 1.1 specification. The device will decode the AC-coupled differential signal of the high-speed eDP in RGB format and drive the TFT panel integrated with the LVDS interface. The built-in LVDS transmitter is equipped with flexible OpenLDI/SPWG bit mapping and is capable of driving single-port or dual-port (18/24-bit) LVDS panels. The LVDS output of the CS5211 can be configured to support resolutions up to 1920x1200 with a refresh rate of 60 Hz. In addition, the CS5211 also has enhanced display features such as backlight inverter control, brightness adjustment, image jitter algorithm and EMI suppression mechanism, which is beneficial to system manufacturers.

The CS5211 is configured as an eDP to LVDS converter; the LVDS transmitter supports single-port 18-bit, single-port 24-bit, dual-port 18-bit, and dual-port 24-bit transmissions, and supports OpenLDI and SPWG bit mapping.

